

Joel Eui Sun Ahn

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Portfolio: <http://bit.ly/2A2S5In>

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EDUCATION

- **University of British Columbia** Sep. 2015 – Present
BaSc.in Engineering Physics GPA: 83.4% — 3.8/4.33 Expected Graduation Date: May 2021

WORK EXPERIENCE

- **Qumulo** Vancouver, BC
Software Engineer May – Aug 2019
 - Helped implement in **C** data Snapshot Replication and retention across distributed file systems, working both back-end to interface with file system APIs and creating **Typescript** Web UIs for convenient usage
 - Improved Replication progress reporting using a more robust algorithm based on a moving average of data stream throughput and adding greater details to progress UI
 - Improved backwards compatibility testing of Snapshot Replication feature using **Python** scripts to automatically generate JSON configuration of versions to test against
- **Intel of Canada, Ltd.** Vancouver, BC
Firmware Engineer May – Dec 2018
 - SSD Media PHY Driver: Assisted engineers validate new product boards by rapidly developing firmware to initialize and validate PHY drivers to NAND/DRAM memory, such as margins training and dynamic PLL changing, enabling them to analyze waveforms and verify signal integrity
 - Electrical Validation Automation: Helped extend Python scripts that helps visualize bit error rate of NAND/DRAM PHY margins training at different delay taps of the read/write data strobe
- **OSI Maritime Systems** Burnaby, BC
Software Engineer Jan – Apr 2017
 - Created **Powershell** scripts that could interface with multiple machines and asynchronously deploy latest company builds remotely
 - Developed **C++** dev tool that serializes company's configuration files into Dewey extended encoding, stores entries in a **SQLite** database for easy editing, and then reconverts to format parseable by company software

TECHNICAL PROJECTS

- **ECDSA Verification ASIC**
Capstone Jun 2019 - Present
 - Designing in **System Verilog** and **C** an ASIC that can perform +100,000 elliptic curve point digital signatures
 - Scripted in **Python** code that could auto-generate Verilog for complex arithmetic logic designs such as Dadda Tree Multiplier and Carry-Propagate Adders
- **UBC Formula Electric**
Firmware Team Sep 2018 - Dec 2018
 - Refactored and tested Power Distribution Module firmware which is responsible for enforcing safety protocols when voltage/currents exceed safe limits
 - Incorporated **CI** to check code compilation using **CMake**; enforce coding standards using clang-format; and validate code using hardware in the loop tests
- **Slinger** Personal Project
Code: <http://bit.ly/2ErED4d> *Video:* <http://bit.ly/2pXKvyI> Mar – Apr 2017
 - Developed in **Java** an Android game wherein a player attempts to climb as high as possible by slinging their avatar from anchor to anchor, avoiding obstacles along the way!

INTERESTS

Sailing — Swimming — Badminton — Baduk — Microcontrollers — Learning new languages